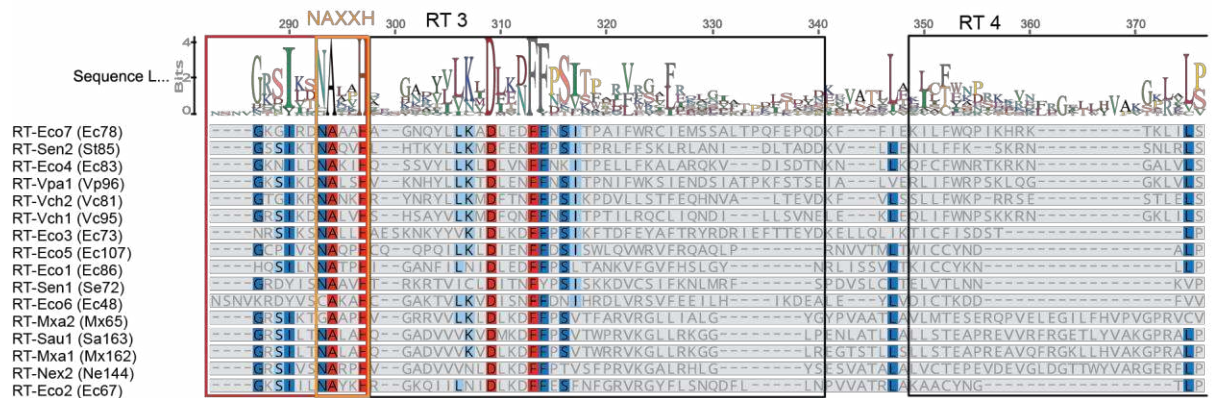
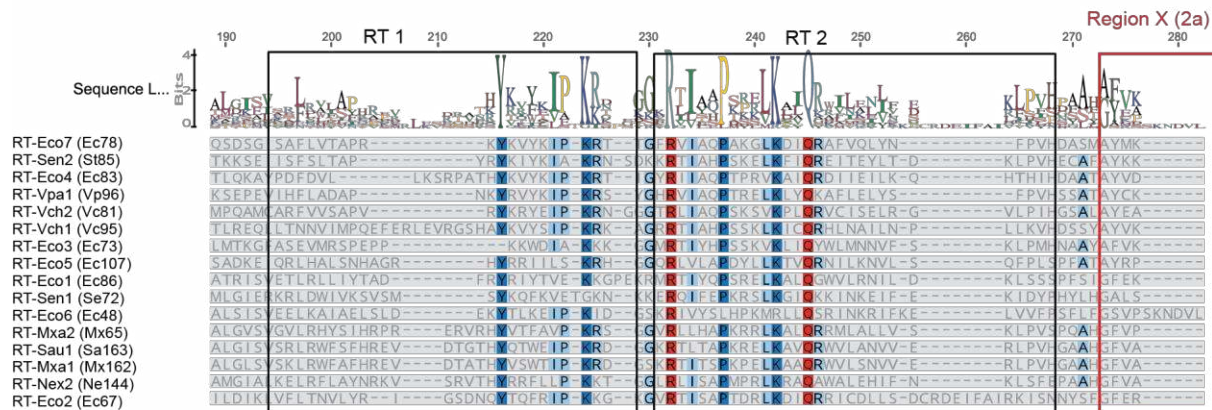
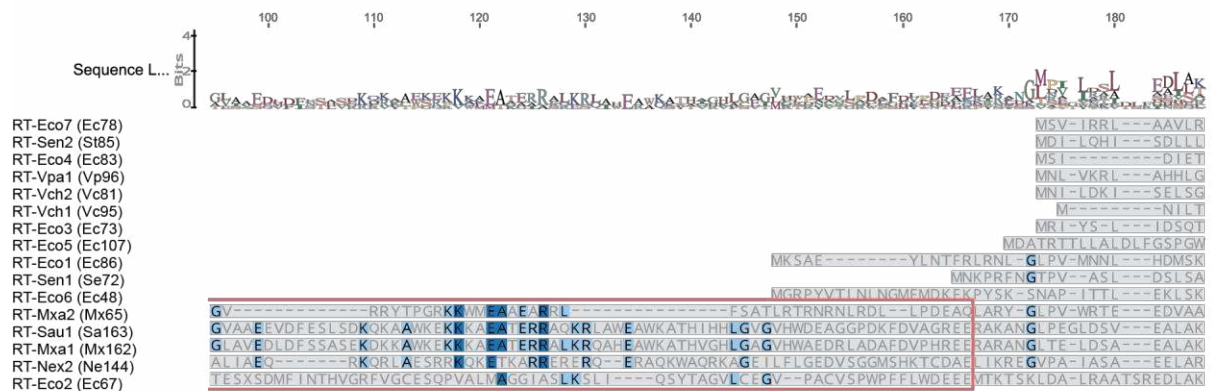
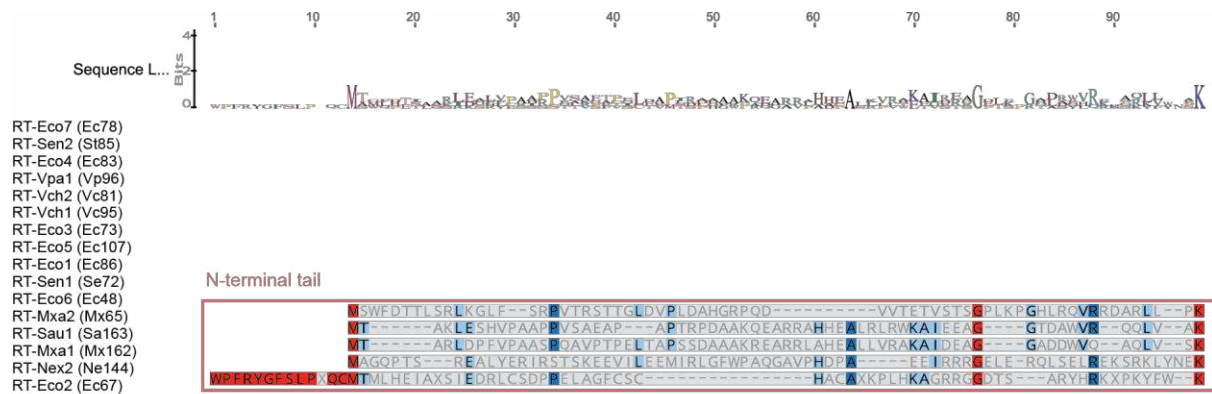




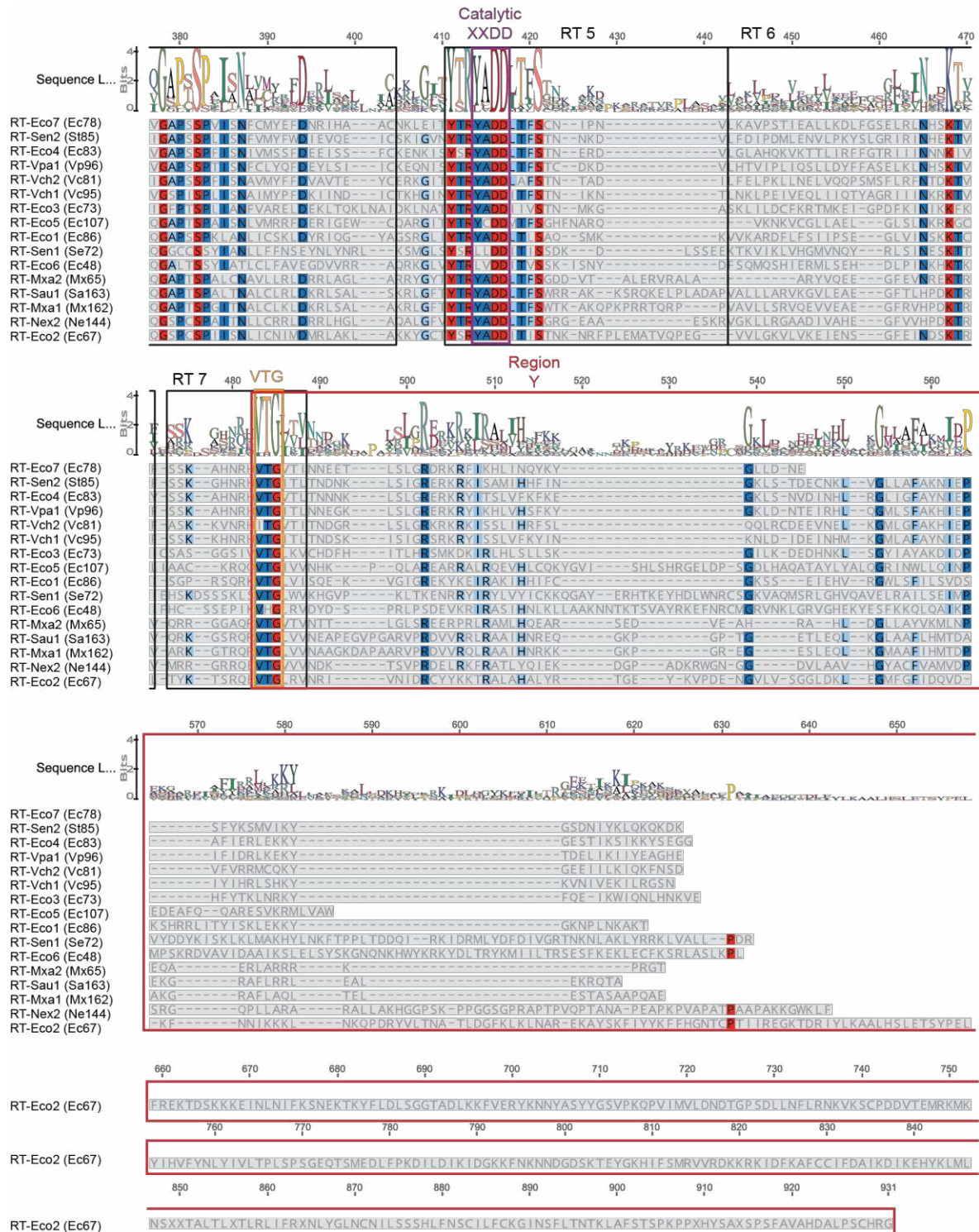


Figure S2. Alignment of all 16 experimentally validated retron RT sequences. Conserved RT regions are boxed in black, retron specific-regions in red, retron-specific motifs NAXXH and VTG in orange, and the catalytic XXDD tetrad in purple.

	Eco7 (Ec78)	Sen2 (St85)	Eco4 (Ec83)	Vpa1 (Vp96)	Vch2 (Vc81)	Vch1 (Vc95)	Eco3 (Ec73)	Eco5 (Ec107)	Eco1 (Ec86)	Sen1 (Se72)	Ec4 (Ec48)	Mxa2 (Mx65)	Sau1 (Se163)	Mxa1 (Mx162)	Nex2 (Ne144)	Eco2 (Ec67)
Eco7 (Ec78)		43.8	42.0	58.2	42.0	42.0	27.7	24.8	23.8	15.4	15.9	21.2	22.3	23.8	19.7	21.8
Sen2 (St85)	43.8		42.9	43.4	49.2	45.5	31.7	22.3	28.3	18.0	14.3	21.3	22.1	23.5	16.8	20.4
Eco4 (Ec83)	42.0	42.9		45.8	39.8	45.7	27.0	22.8	23.4	16.0	13.5	20.1	20.3	21.4	18.2	18.9
Vpa1 (Vp96)	58.2	43.4	45.8		41.2	43.3	26.6	23.6	24.5	16.9	13.7	21.8	20.9	22.8	16.9	17.8
Vch2 (Vc81)	42.0	49.2	39.8	41.2		39.9	28.4	22.2	22.0	16.1	14.8	21.0	21.5	24.0	16.7	18.6
Vch1 (Vc95)	42.0	45.5	45.7	43.3	39.9		27.2	19.7	23.1	16.1	13.9	20.7	20.7	21.4	16.5	19.5
Eco3 (Ec73)	27.7	31.7	27.0	26.6	28.4	27.2		21.3	22.8	14.5	15.1	21.6	22.3	22.0	20.6	18.7
Eco5 (Ec107)	24.8	22.3	22.8	23.6	22.2	19.7	21.3		24.0	22.0	19.6	24.4	25.7	23.0	24.5	22.2
Eco1 (Ec86)	23.8	28.3	23.4	24.5	22.0	23.1	22.8	24.0		20.4	18.1	24.7	24.8	26.2	25.4	25.8
Sen1 (Se72)	15.4	18.0	16.0	16.9	16.1	16.1	14.5	22.0	20.4		25.1	17.5	17.7	17.5	19.2	19.6
Ec6 (Ec48)	15.9	14.3	13.5	13.7	14.8	13.9	15.1	19.6	18.1	25.1		15.7	16.9	18.4	16.2	16.6
Mxa2 (Mx65)	21.2	21.3	20.1	21.8	21.0	20.7	21.6	24.4	24.7	17.5	15.7		34.5	33.3	31.4	19.9
Sau1 (Se163)	22.3	22.1	20.3	20.9	21.5	20.7	22.3	25.7	24.8	17.7	16.9	34.5		73.0	35.4	23.4
Mxa1 (Mx162)	23.8	23.5	21.4	22.8	24.0	21.4	22.0	23.0	26.2	17.5	18.4	33.3	73.0		34.5	23.3
Nex2 (Ne144)	19.7	16.8	18.2	16.9	16.7	16.5	20.6	24.5	25.4	19.2	16.2	31.4	35.4	34.5		21.7
Eco2 (Ec67)	21.8	20.4	18.9	17.8	18.6	19.5	18.7	22.2	25.8	19.6	16.6	19.9	23.4	23.3	21.7	

Table S1. Comparison of retron RT sequence identity. Identity was compared by aligning all RT sequences in NCBI's online Constrain-based Multiple Alignment Tool (COBALT) (2), with the following parameters: gap penalty opening: -11; gap penalty extension: -1; end gap opening: -5; end gap extension: -1; RPS BLAST to guide alignment on; constrain E-value 0.003; find conserved columns and recompute alignment on; use query clustering parameters on; word size 4; max cluster distance 0.8; and alphabet regular.

REFERENCES

1. Zuker, M. (2003) Mfold web server for nucleic acid folding and hybridization prediction. *Nucleic Acids Research*, **31**, 3406-3415.
2. Papadopoulos, J.S. and Agarwala, R. (2007) COBALT: constraint-based alignment tool for multiple protein sequences. *Bioinformatics*, **23**, 1073-1079.